02 · Topics: what are the calls about? (Task 1)

Transcript Intelligence: notebook 2 of 5

The task. Build a pipeline that categorizes calls by topic or theme, explain why the approach was chosen, and give examples from each category.

The answer in one line. A **hybrid** pipeline (`src/topics.py`): unsupervised clustering *discovers* the structure, a fixed 10-theme taxonomy *defines* it in business language, Claude *classifies* each call (one primary theme plus every theme that applies), and keyword rules provide both a fallback and an independent check.

How to run. `python -m src.pipeline` builds two versions of the topic outputs: `data/processed/hybrid/` (Claude + rules) and `data/processed/rules/` (keyword rules only, no API calls). This notebook reads both and compares them. It makes no LLM calls.

1. Setup

In [1]:
```
%load_ext autoreload
%autoreload 2

import sys, json
from pathlib import Path

ROOT = Path.cwd().parent if Path.cwd().name == "notebooks" else Path.cwd()
sys.path.insert(0, str(ROOT))

import numpy as np
import pandas as pd
import plotly.express as px
import plotly.io as pio

from src.load import PROCESSED_DIR, mode_dir
from src.enrich import load_enriched, PERIODS
from src.topics import THEMES, THEME_NAMES, SYSTEM_PROMPT, N_CLUSTERS, EMBED_MODEL, load_topics, call_text, embed_calls
```

```
from src.llm import load_usage, uncosted_cache_entries

pio.renderers.default = "plotly_mimetype+notebook_connected"
pd.set_option("display.max_columns", 50)
pd.set_option("display.max_colwidth", 140)
TYPE_COLORS = {"external": "#2a78d6", "support": "#eb6834", "internal": "#8a8f98"}
```

In [2]:
```
calls = load_enriched(PROCESSED_DIR)["calls"]
ORIGINAL_END = pd.Timestamp("2026-04-28").date()
calls["is_synthetic"] = calls["date"].map(lambda d: pd.Timestamp(d).date() > ORIGINAL_END)

hyb, clusters, hyb_metrics = load_topics(mode_dir("hybrid"))
rul, _, rul_metrics = load_topics(mode_dir("rules"))

meta = ["call_id", "date", "title", "call_type", "account", "period", "sentiment_score", "is_synthetic"]
H = hyb.merge(calls[meta], on="call_id")
R = rul.merge(calls[meta], on="call_id")
Ho = H[~H["is_synthetic"]]          # original 100 calls, used for headline numbers
print(f"{len(H)} calls classified ({len(Ho)} original, {H['is_synthetic'].sum()} synthetic); "
      f"hybrid sources: {hyb['source'].value_counts().to_dict()}; model: {hyb_metrics.get('model')}")
```

```
115 calls classified (100 original, 15 synthetic); hybrid sources: {'llm': 115}; model: claude-sonnet-5-5
```

2. Why not just use the provided topics?

Each call already has free-text `topics` from the transcription vendor. `01_explore` showed they can't be used directly: 351 distinct labels across 100 calls, most used only once, and one concept split many ways ("outage", "platform outage", "service outage", "incident response"...). A stakeholder can't filter or trend on that. We need a **small, stable, readable** set of categories.

Options considered

Approach	Strength	Weakness	Role here
Keyword rules	Free, instant, fully transparent	Brittle: matches words, not meaning; misses context	Fallback + independent check
Clustering (embeddings)	Finds structure without assumptions	Clusters aren't named, can form around noise (a customer's name)	**Discover** the themes

Approach	Strength	Weakness	Role here
LLM classification	Understands context, handles calls with several themes, explains itself	Costs money; non-deterministic unless controlled	**Classify** each call
Hybrid (chosen)	Data-grounded taxonomy + nuanced labels + a built-in cross-check	More moving parts	✓

Decision. Use each method for what it's good at, and use the rules to *check* the LLM rather than trust it blindly.

3. Step 1: Discover (clustering)

Each call's title, summary and vendor topics are embedded with a local sentence-embedding model (`all-MiniLM-L6-v2` , free, no API) and grouped with KMeans into 10 clusters. Each cluster is described by its most distinctive TF-IDF terms.

In [3]:
```
clusters.sort_values("n_calls", ascending=False)[["cluster", "n_calls", "top_terms", "dominant_theme", "purity"]]
```

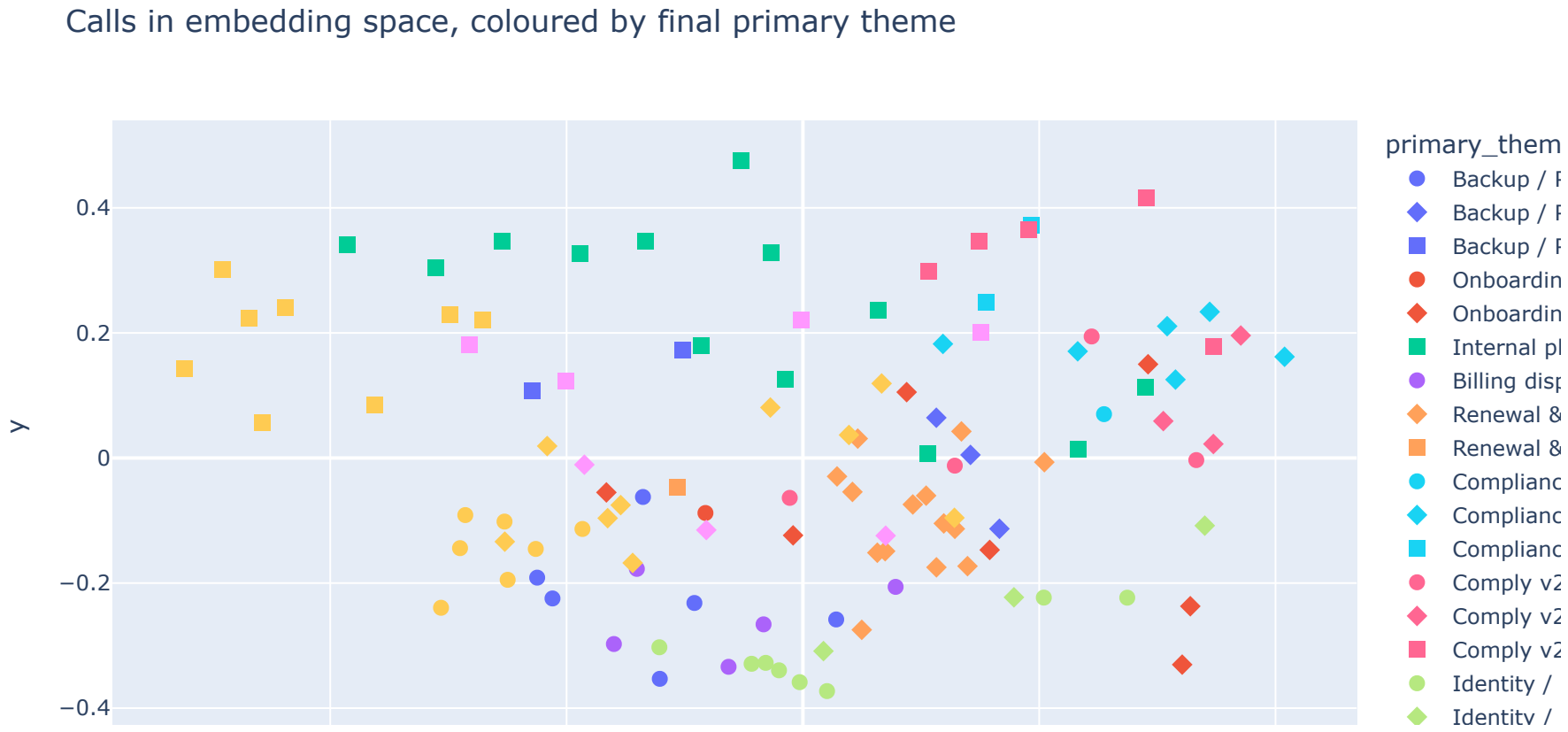
Out[3]:

	cluster	n_calls	top_terms	dominant_theme	purity
3	3	20	outage, failure, incident, detect, event, communication	Detect outage & fallout	0.75
4	4	17	renewal, pricing, aegis, outage, competitive, review	Renewal & contracts	0.65
2	2	16	hipaa, v2, comply, reporting, comply v2, pci dss	Comply v2 launch	0.56
6	6	11	mfa, sync, identity, policy, derek, sso	Identity / SSO / MFA	0.45
1	1	10	kendra, brightpath, alert, tuning, false, detection	Detect outage & fallout	0.30
0	0	10	post, post mortem, mortem, team, outage, incident	Backup / Protect	0.30
7	7	10	backup, vanta, cloudprime, data, connector, hybrid	Backup / Protect	0.60
8	8	8	billing, overage, invoice, dispute, usage, discrepancy	Billing disputes	0.62
9	9	7	request, feature, role, victor, frostbyte, custom	Identity / SSO / MFA	0.43
5	5	6	sprint, sprint planning, team, planning, pipeline, retrospective	Internal planning	0.83

In [4]:
```python
# 2-D map of the clusters (PCA of the same embeddings used for clustering)
from sklearn.decomposition import PCA
texts = calls.set_index("call_id").loc[H["call_id"]].reset_index().apply(call_text, axis=1).tolist()
xy = PCA(n_components=2, random_state=0).fit_transform(embed_calls(texts))
plot = H.assign(x=xy[:, 0], y=xy[:, 1], cluster=H["cluster"].astype(str))
px.scatter(plot, x="x", y="y", color="primary_theme", symbol="call_type", hover_data=["title", "cluster"],
           title="Calls in embedding space, coloured by final primary theme").update_traces(marker_size=9)
```

Loading weights: 100% 103/103 [00:00<00:00, 2344.48it/s]

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
```



What this shows. Most clusters map onto a recognisable theme: the outage (two clusters), Comply v2, renewals, billing, backup, Identity and sprint planning. That confirms these themes are real in the data, not invented.

It also shows why clustering **can't be the final answer**:

- **Some clusters form partly around people, not topics.** In the current run one cluster's top terms include a customer name ("silverline"), and two others include a staff member's first name ("maria", "marcus"). KMeans groups whatever is most similar, and repeated names are

very similar. The exact terms shift a little when new transcripts are added.

- **Each call gets exactly one cluster**, but many calls are about several things. A renewal call dominated by the outage belongs in both groups.
- **Clusters have no names.** Someone has to turn "renewal, aegis, pricing, outage, maria" into a category a sales leader would use.

4. Step 2: Define the taxonomy

The 10 themes were consolidated from the clusters above, the vendor's 351 labels, and the call types. Some business-critical themes, such as **Billing disputes** and **Competitive threat**, are too small or too spread out to form their own cluster, but a stakeholder still needs them as categories. Each theme has a plain-English **definition** (what the LLM sees) and a **keyword rule** (the fallback and check).

In [5]:
```
pd.DataFrame([{"theme": n, "definition": d, "keyword rule": p.pattern} for n, (d, p) in THEMES.items()])
```

Out[5]:

	theme	definition	keyword rule
0	Compliance & audit	Regulatory or security compliance work: SOC 2, PCI DSS, HIPAA, ISO 27001, audits, compliance reporting and evidence.	complian|audit|SOC ?2|PCI|HIPAA|ISO ?27001|GDPR
1	Detect outage & fallout	The March Detect pipeline outage and its consequences: war rooms, escalations, lost threat visibility, data gaps, root cause, remediatio...	outage|war room|post-incident|incident|root cause|pipeline failure|loss of threat visibility|dashboard down|alerts not firing|data gaps|...
2	Comply v2 launch	The Comply v2 release: design, launch readiness, GA (7 Apr), previews, early access, deployment of v2 and adoption.	Comply v2|Comply 2|early access|launch readiness|\bGA\b
3	Renewal & contracts	Commercial conversations: renewals, contract terms, pricing, discounts, multi-year deals, expansions and account reviews with a commerci...	renewal|contract|multi-year|pricing|discount|expansion|upsell
4	Internal planning	AegisCloud-only operating meetings: sprint planning, retros, standups, roadmap, all-hands, quarterly planning, internal strategy reviews.	sprint|standup|stand-up|all hands|retro|roadmap|quarterly planning|win/loss
5	Competitive threat	Competitors come up as a real risk: the customer is evaluating, comparing or considering switching to another vendor (SentinelShield, Va...	competit|vendor comparison|SentinelShield|VaultEdge|CyberNova|Fortiguard|switching
6	Identity / SSO / MFA	The Identity product and access management: SSO, MFA, SAML, SCIM, LDAP, provisioning, group policy sync, login failures.	\bIdentity\b|SSO|MFA|SAML|SCIM|LDAP|provisioning|login
7	Backup / Protect	The Protect product and data protection: backups, restores, snapshots, backup windows and backup performance.	\bProtect\b|backup|restore|snapshot
8	Onboarding & deployment	Getting a customer live: onboarding kickoffs, module setup, deployment, configuration, integration rollout.	onboarding|kickoff|deployment|setup|implementation|rollout

	theme	definition	keyword rule
9	Billing disputes	Invoices, billing errors, overage charges, seat-count discrepancies, refunds or credits being disputed.	invoice|billing|overage|discrepanc|refund|charges? dispute

5. Step 3: Classify each call with Claude

For each call, Claude receives the title, call type, date, summary, vendor topics and key moments, and returns structured JSON validated by a pydantic schema:

Field	Meaning
`primary_theme`	the single theme the call is mostly about
`themes`	every theme that is a substantive part of the call (multi-label)
`certainty`	confidence in the primary theme, 0–1
`rationale`	one sentence citing the evidence

Engineering choices

- **Constrained output.** Themes must come from the taxonomy, so the model can't invent a category.
- **Cached.** Each answer is stored in `data/processed/llm_cache/`, keyed by model, prompt version and input. Re-runs are free and give identical results. Bump `PROMPT_VERSION` to invalidate.
- **Fails soft.** If the API is unavailable or the model declines, keyword rules classify that call and it's marked `source = "rules"`.
- **Explainable.** Every label carries a rationale a reviewer can check.
- **Business calls only.** Non-business calls (birthdays, farewells, socials) are screened out before classification (`01_explore`, section 4). That's why the taxonomy needs no *Social / other* theme: Claude never has to force a birthday call into a business category, and the themes' counts aren't padded by calls that aren't about work.

The full instructions sent to the model:

In [6]:
```
print(SYSTEM_PROMPT)
```

```
You categorize recorded calls at AegisCloud, a B2B cloud-security company whose products are Detect, Protect, Comply, Identity
and LogVault. Assign each call to themes from this fixed taxonomy:

- Compliance & audit: Regulatory or security compliance work: SOC 2, PCI DSS, HIPAA, ISO 27001, audits, compliance reporting an
d evidence.
- Detect outage & fallout: The March Detect pipeline outage and its consequences: war rooms, escalations, lost threat visibilit
y, data gaps, root cause, remediation, SLA credits, trust repair.
- Comply v2 launch: The Comply v2 release: design, launch readiness, GA (7 Apr), previews, early access, deployment of v2 and a
doption.
- Renewal & contracts: Commercial conversations: renewals, contract terms, pricing, discounts, multi-year deals, expansions and
account reviews with a commercial outcome.
- Internal planning: AegisCloud-only operating meetings: sprint planning, retros, standups, roadmap, all-hands, quarterly plann
ing, internal strategy reviews.
- Competitive threat: Competitors come up as a real risk: the customer is evaluating, comparing or considering switching to ano
ther vendor (SentinelShield, VaultEdge, CyberNova, Fortiguard Pro), or AegisCloud is planning its competitive response.
- Identity / SSO / MFA: The Identity product and access management: SSO, MFA, SAML, SCIM, LDAP, provisioning, group policy syn
c, login failures.
- Backup / Protect: The Protect product and data protection: backups, restores, snapshots, backup windows and backup performanc
e.
- Onboarding & deployment: Getting a customer live: onboarding kickoffs, module setup, deployment, configuration, integration r
ollout.
- Billing disputes: Invoices, billing errors, overage charges, seat-count discrepancies, refunds or credits being disputed.

Rules:
- primary_theme is the single theme the call is mostly about.
- themes lists every theme that is a substantive part of the call, including the primary. Leave out themes that only come up in
passing.
- certainty is your confidence in primary_theme, from 0 to 1.
- rationale is one short sentence that cites what in the call supports the primary theme.
```

In [7]:
```
usage = load_usage()
topics_usage = usage[usage["step"] == "topics"]
print(f"Cached topic answers on disk: {len(list((PROCESSED_DIR / 'llm_cache').glob('topics_*.json')))}")
print(f"Topic API calls in the cost log: {len(topics_usage)}, total ${topics_usage['cost_usd'].sum():.4f} "
      f"(avg ${topics_usage['cost_usd'].mean():.4f} per call)" if len(topics_usage) else "No topic calls logged yet")
print(f"Cached answers from before cost logging existed: {uncosted_cache_entries()}")
```

```
Cached topic answers on disk: 115
Topic API calls in the cost log: 15, total $0.0804 (avg $0.0054 per call)
Cached answers from before cost logging existed: 108
```

6. Results: the categories

Counts below are for the **original 100 calls**.

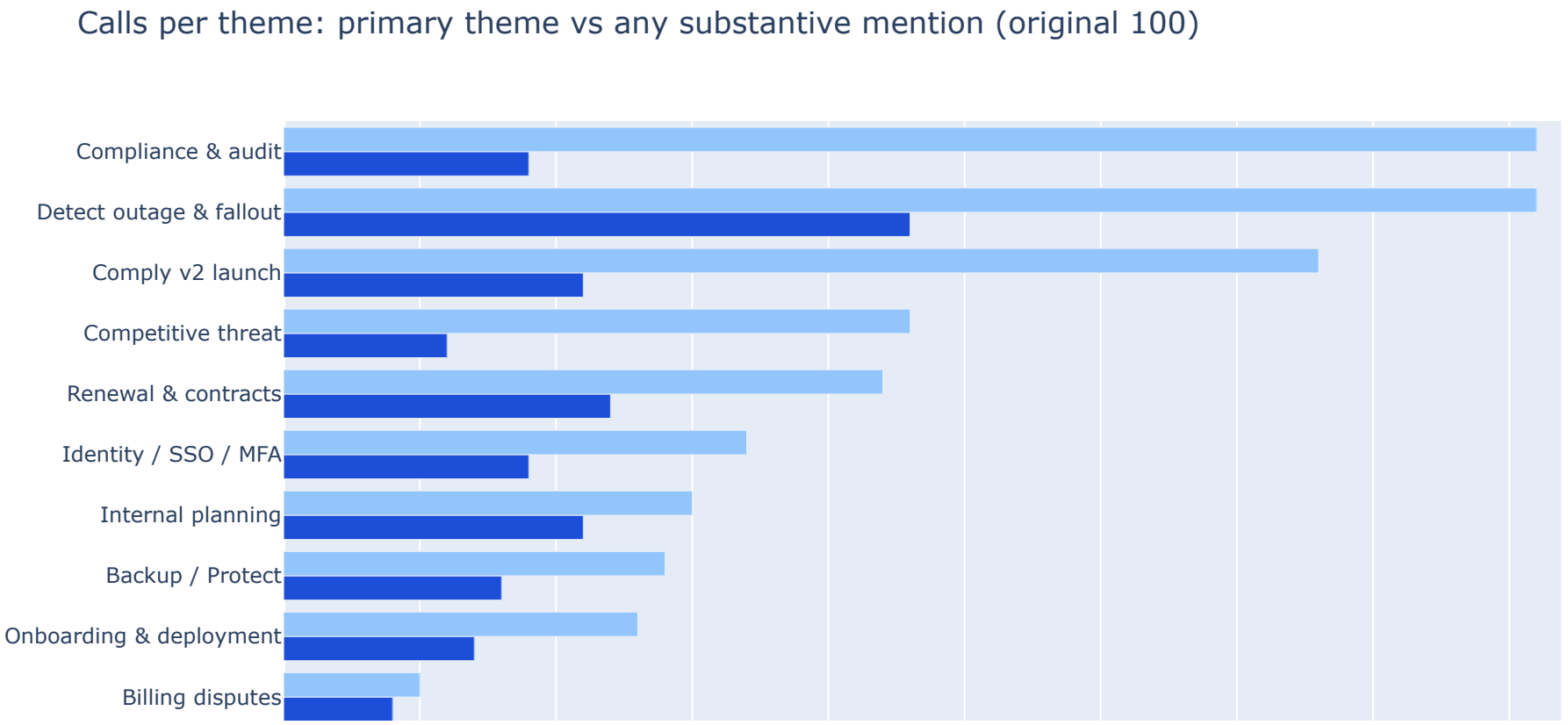
In [8]:
```
prim = Ho["primary_theme"].value_counts().rename("primary")
multi = Ho["themes"].map(list).explode().value_counts().rename("any mention")
tbl = pd.concat([prim, multi], axis=1).fillna(0).astype(int).reindex(THEME_NAMES)
tbl["share_primary"] = (tbl["primary"] / len(Ho)).round(2)
tbl.sort_values("any mention", ascending=False)
```

Out[8]:

	primary	any mention	share_primary
Compliance & audit	9	46	0.09
Detect outage & fallout	23	46	0.23
Comply v2 launch	11	38	0.11
Competitive threat	6	23	0.06
Renewal & contracts	12	22	0.12
Identity / SSO / MFA	9	17	0.09
Internal planning	11	15	0.11
Backup / Protect	8	14	0.08
Onboarding & deployment	7	13	0.07
Billing disputes	4	5	0.04

In [9]:
```
long = tbl.reset_index(names="theme").melt(id_vars="theme", value_vars=["primary", "any mention"],
                                           var_name="count", value_name="calls")
fig = px.bar(long, y="theme", x="calls", color="count", barmode="group", orientation="h",
             category_orders={"theme": tbl.sort_values("any mention").index.tolist()[::-1]},
```

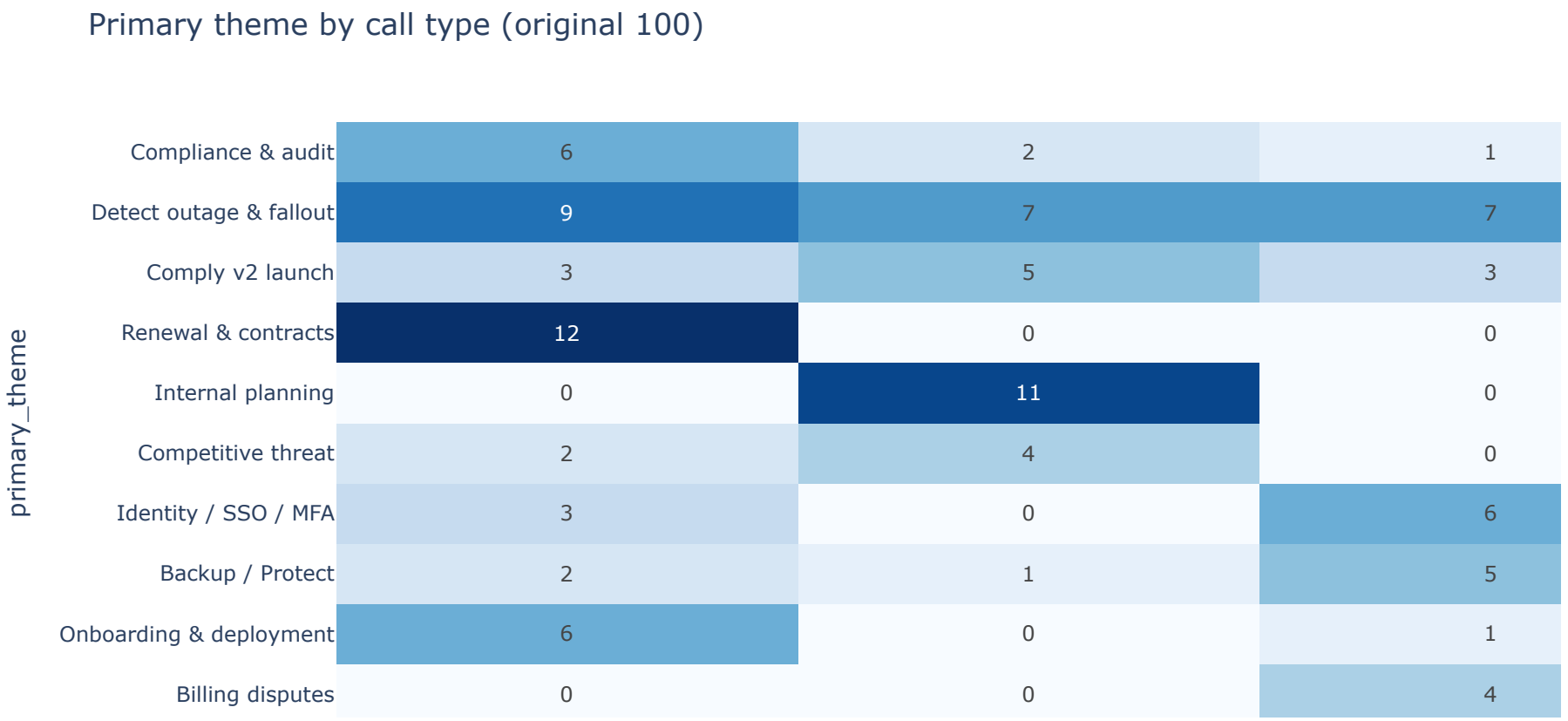
```
            color_discrete_map={"primary": "#1D4ED8", "any mention": "#93C5FD"},
            title="Calls per theme: primary theme vs any substantive mention (original 100)")
fig.update_layout(yaxis_title=None)
fig
```



In [10]:
```
print("Themes per call:", Ho["themes"].map(len).value_counts().sort_index().to_dict())
print(f"Calls with 2+ themes: {(Ho['themes'].map(len) >= 2).mean():.0%}")
```

```
Themes per call: {1: 26, 2: 32, 3: 24, 4: 13, 5: 5}
Calls with 2+ themes: 74%
```

In [11]:
```
heat = pd.crosstab(Ho["primary_theme"], Ho["call_type"]).reindex(THEME_NAMES).fillna(0).astype(int)
px.imshow(heat, text_auto=True, color_continuous_scale="Blues", aspect="auto",
          title="Primary theme by call type (original 100)").update_layout(coloraxis_showscale=False)
```



What this shows.

- **The Detect outage is the single largest theme** and appears in every call type: internal war rooms, customer escalations and support tickets. No other theme cuts across the company like that.
- **Compliance is the theme people talk about, not the one calls are about.** It appears in about half of all calls, but is rarely the *primary* theme. It's the context (customers are regulated) rather than the purpose of the call. This is why multi-label matters: a single-label classifier would make compliance look minor.
- **Most calls are about more than one thing.** That supports the decision to keep every substantive theme rather than force one label.
- **Each call type has a clear profile.** Renewals are external, Identity and Backup problems arrive through support, and planning is internal. That's expected, which gives confidence the labels are sensible.

7. Examples from each category

The assignment asks for examples. For each theme, the two most confident calls from the original data, with Claude's one-line reasoning.

In [12]:
```
ex = (Ho.sort_values("certainty", ascending=False)
        .groupby("primary_theme", group_keys=False).head(2)
        .assign(theme=lambda d: pd.Categorical(d["primary_theme"], THEME_NAMES, ordered=True))
        .sort_values(["theme", "certainty"], ascending=[True, False]))
ex[["theme", "call_type", "title", "certainty", "rationale"]].reset_index(drop=True)
```

Out[12]:

	theme	call_type	title	certainty	rationale
0	Compliance & audit	internal	SOC 2 Type II - Final Review	0.93	The call is a final walk-through of SOC 2 Type II audit materials, covering controls, evidence packaging, and the submission and audit w...
1	Compliance & audit	external	Aegis / Maplewood Goods - SOC 2 Prep Review	0.90	The call centers on Maplewood's SOC 2 audit prep, manual evidence collection pain, and multi-framework control mapping (SOC 2, PCI DSS, ...
2	Detect outage & fallout	internal	INCIDENT: Detect Pipeline Failure - War Room	0.98	The call is an internal war room on the Detect pipeline failure, covering root cause (single-node ingestion), customer escalations from ...
3	Detect outage & fallout	support	Support Case #6573 - Ridgeline Logistics Detect Data Gaps	0.97	Customer reports a total Detect dashboard outage from the ingestion pipeline failure, with lost OT/SCADA visibility, Sev 1 escalation, d...
4	Comply v2 launch	internal	Comply v2 - GA Deployment Plan	0.96	Internal pre-launch alignment for the Comply v2 GA on April 7 covering staging dry-runs, regression testing, phased rollout, and rollbac...
5	Comply v2 launch	internal	Comply v2 - Launch Day Checklist	0.96	Internal pre-launch sync reviewing QA bug status, HIPAA module validation, onboarding flows and readiness for the April 7 Comply v2 GA.
6	Renewal & contracts	external	Aegis / Axiom Labs - Multi-Year Renewal	0.93	The call is a renewal planning session where the CSM presents a three-year proposal with an elastic seat model and enterprise API tier, ...
7	Renewal & contracts	external	Aegis / Bridgeport Health - Renewal Finalization	0.93	The call is a renewal finalization where flat pricing, free year-one Comply v2 access, and a 20% seat expansion rider are agreed, with t...
8	Internal planning	internal	Detect Team - Sprint Planning	0.96	This is an internal Detect team sprint planning meeting covering carryover tickets, a design spike, and committing to 13 story points.
9	Internal planning	internal	Detect Team - Sprint Planning	0.93	This is an internal Detect team sprint twelve planning meeting with a sprint eleven retro, carry-overs, capacity scoping and sprint comm...

	theme	call_type	title	certainty	rationale
10	Competitive threat	internal	Competitive Response Strategy	0.90	The internal call was convened to plan AegisCloud's response to SentinelShield's aggressive positioning in late-stage deals, including a...
11	Competitive threat	external	Aegis / Brightpath Commerce - Competitive Evaluation	0.90	The call is titled a competitive evaluation and centers on Brightpath's six-week evaluation of SentinelShield, with a mid-May decision d...
12	Identity / SSO / MFA	support	Support Case #3677 - Coastal Living Co MFA Token Failures	0.97	The call is a P1 support case about TOTP validation failures (error 4031) locking out 40-45 users via the Okta SSO integration with Aegi...
13	Identity / SSO / MFA	support	Support Case #6635 - Nova Retail Group SSO Login Failures	0.96	The call is a support case about SSO login failures for about 40% of users caused by a SAML NameID regression in the Identity module, wi...
14	Backup / Protect	support	Support Case #6977 - Brightpath Commerce Slow Backup Performance	0.97	The call is a support case about nightly backup jobs slowing from 90 minutes to 4-6 hours, traced to an outdated backup agent (3.4.1) wi...
15	Backup / Protect	support	Support Case #1514 - Meridian Capital Granular Restore Request	0.96	The call is a support case about missing granular file-level restore from EBS snapshots in Aegis Protect, including a two-hour workaroun...
16	Onboarding & deployment	external	Aegis / Brightpath Commerce - Detect Module Deployment	0.93	The call is an onboarding planning session for the Detect deployment covering sequencing, a one-month timeline, the late-March peak-seas...
17	Onboarding & deployment	external	Aegis / Coastal Living Co - Onboarding Kickoff	0.90	This is an onboarding kickoff where Maria walks through the 4-6 week three-phase plan covering environment discovery, backup policy conf...
18	Billing disputes	support	Support Case #9279 - Summit Trust Billing Inquiry	0.93	The call centers on a January invoice showing 1,400 seats versus 1,200 contracted, traced to leftover test accounts, with a billing adju...
19	Billing disputes	support	Support Case #7572 - Brightpath Commerce Invoice Discrepancy	0.93	The call centers on a $1,400 invoice discrepancy from a 2.31 TB overage charge and a request for a billing adjustment, with the Protect ...

In [13]:
```
# Browse any theme in full: change THEME to any name in THEME_NAMES
THEME = "Billing disputes"
H[H["themes"].map(lambda t: THEME in t)][["date", "call_type", "title", "primary_theme", "certainty"]] \
  .sort_values("date").reset_index(drop=True)
```

Out[13]:

	date	call_type	title	primary_theme	certainty
0	2026-02-06	support	Support Case #9279 - Summit Trust Billing Inquiry	Billing disputes	0.93
1	2026-02-25	external	Aegis / Atlas Precision - Contract Discussion	Renewal & contracts	0.72
2	2026-02-27	support	Support Case #7572 - Brightpath Commerce Invoice Discrepancy	Billing disputes	0.93
3	2026-04-10	support	Support Case #8179 - Clearwater Medical Overage Charges Dispute	Billing disputes	0.93
4	2026-04-18	support	Support Case #7570 - Silverline Brands Comply v2 License Upgrade	Billing disputes	0.90
5	2026-06-09	internal	Pricing & Packaging Review - Comply Usage Tiers	Internal planning	0.55
6	2026-06-11	support	Support Case #7311 - Axiom Labs Elastic Seat Invoice Error	Billing disputes	0.96

8. Validation: should we trust the labels?

There's no ground truth in the data, so the labels are checked three ways.

8a. LLM vs keyword rules

The rules were written independently of the LLM, so agreement is evidence that both are capturing something real, and disagreement shows where judgement was needed.

In [14]:
```
cmp = H[["call_id", "title", "primary_theme", "rationale"]].merge(
    R[["call_id", "primary_theme"]].rename(columns={"primary_theme": "rules_primary"}), on="call_id")
agree = (cmp["primary_theme"] == cmp["rules_primary"]).mean()
pd.Series({
    "primary theme: LLM = rules": f"{agree:.0%}",
    "LLM primary appears anywhere in the rule themes": f"{hyb_metrics['primary_in_rule_themes']:.0%}",
    "theme-set overlap, LLM vs rules (mean Jaccard)": hyb_metrics["mean_jaccard_llm_vs_rules"],
    "cluster purity, hybrid labels (weighted)": hyb_metrics["cluster_purity_weighted"],
```

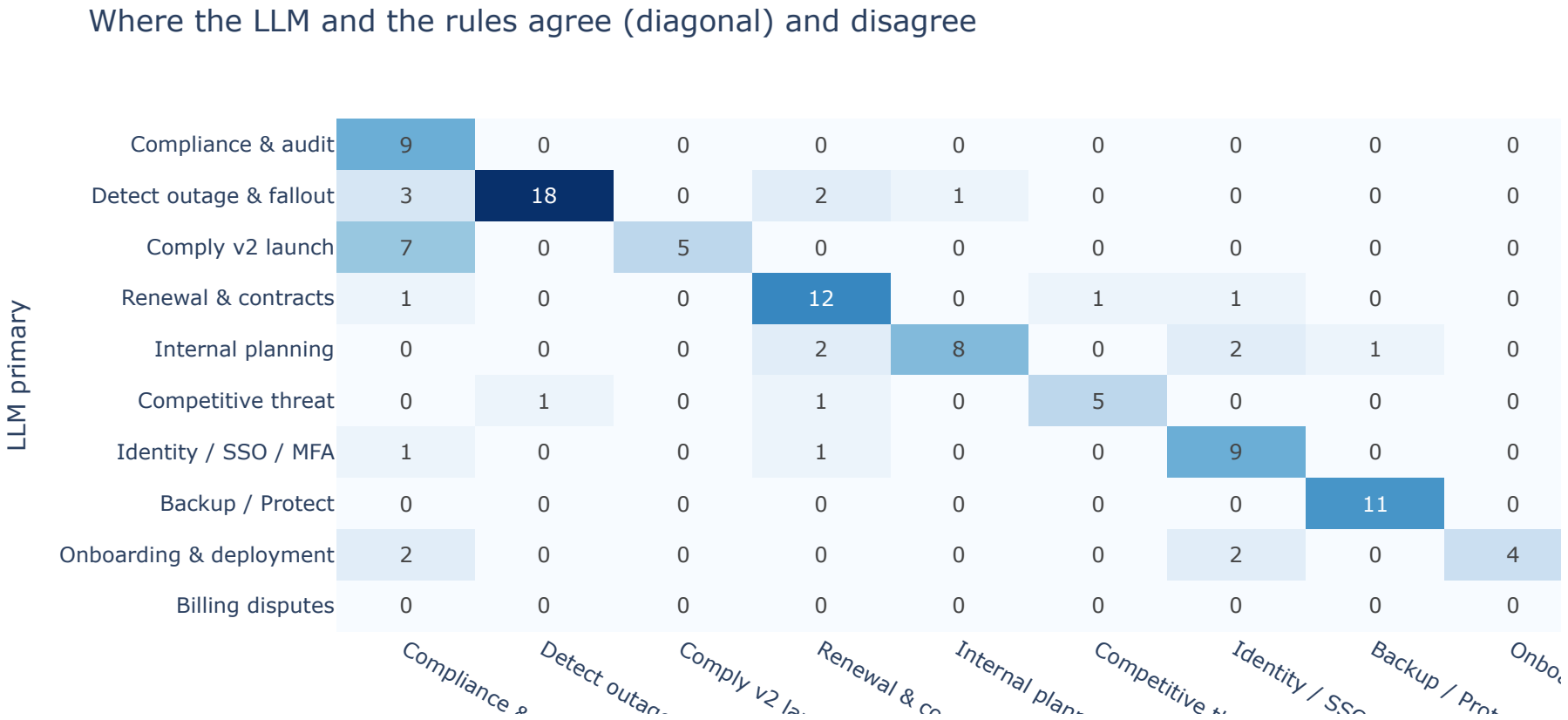
```
    "cluster purity, rules labels (weighted)": rul_metrics["cluster_purity_weighted"],
}, name="value").to_frame()
```

Out[14]:

	value
primary theme: LLM = rules	75%
LLM primary appears anywhere in the rule themes	96%
theme-set overlap, LLM vs rules (mean Jaccard)	0.682
cluster purity, hybrid labels (weighted)	0.564
cluster purity, rules labels (weighted)	0.591

In [15]:

```
ct = pd.crosstab(cmp["primary_theme"], cmp["rules_primary"]).reindex(index=THEME_NAMES, columns=THEME_NAMES).fillna(0).astype(
px.imshow(ct, text_auto=True, color_continuous_scale="Blues", aspect="auto",
          labels={"x": "rules primary", "y": "LLM primary"},
          title="Where the LLM and the rules agree (diagonal) and disagree").update_layout(coloraxis_showscale=False)
```



In [16]:
```
dis = cmp[cmp["primary_theme"] != cmp["rules_primary"]]
print(f"{len(dis)} disagreements. Most common:")
print(dis.groupby(["rules_primary", "primary_theme"]).size().sort_values(ascending=False).head(5).to_string())
dis[["title", "rules_primary", "primary_theme", "rationale"]].head(8).reset_index(drop=True)
```

```
29 disagreements. Most common:
rules_primary         primary_theme
Compliance & audit    Comply v2 launch           7
                      Detect outage & fallout    3
                      Onboarding & deployment    2
Identity / SSO / MFA  Internal planning          2
Renewal & contracts   Detect outage & fallout    2
```

Out[16]:

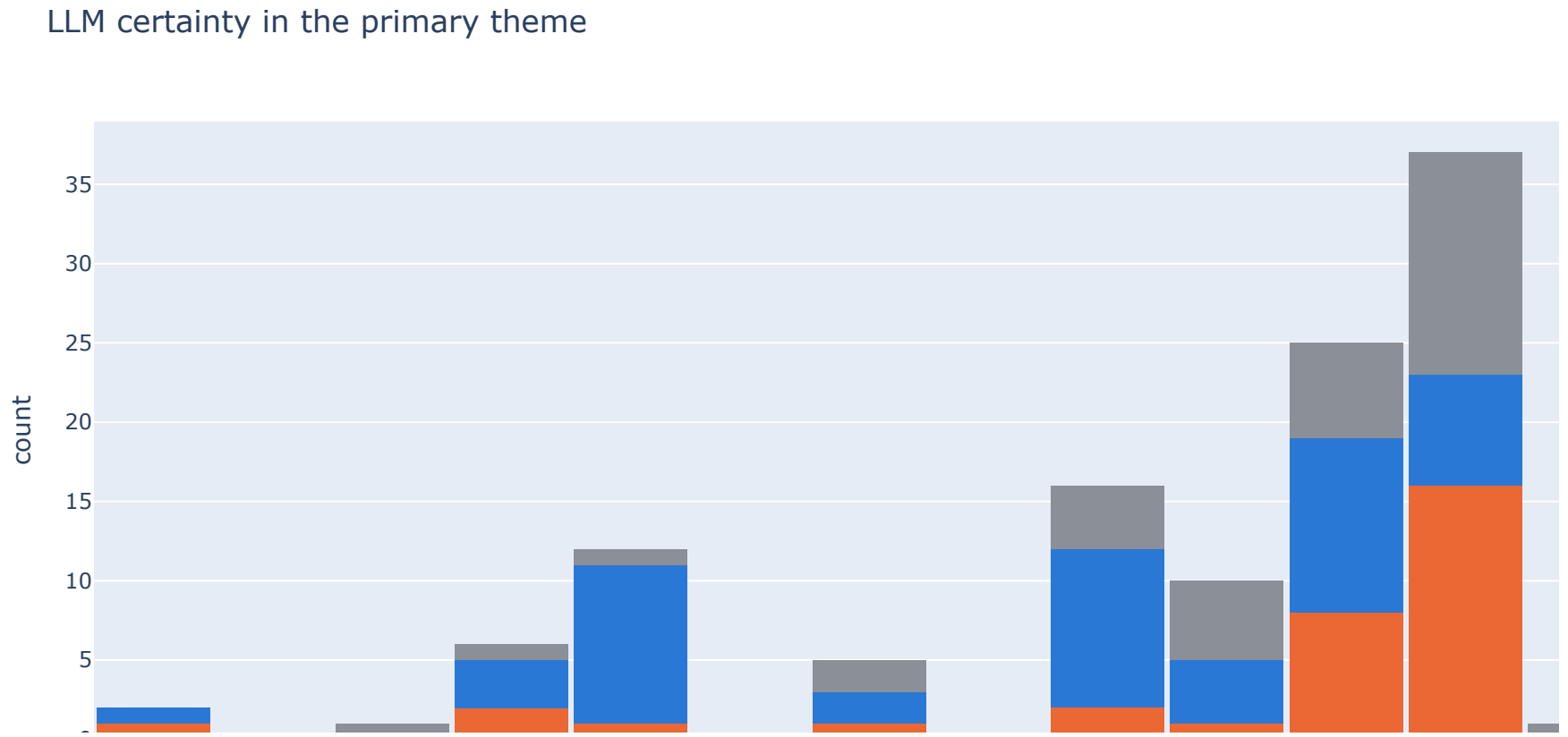
	title	rules_primary	primary_theme	rationale
0	Product Sync - Identity Roadmap	Identity / SSO / MFA	Internal planning	This is an internal product sync on the Identity Q2 roadmap, covering sprint planning, spec-lock and QA process changes, with MFA rules ...
1	Comply v2 - Design Review	Compliance & audit	Comply v2 launch	Internal design review of Comply v2 covering async report generation, unified evidence store, PDF export scope, v2.1 deferral and load t...
2	Support Case #1750 - Vanta Health Systems SIEM Connector Issue	Compliance & audit	Onboarding & deployment	The call is a support case about a LogVault SIEM connector 3.4.1 bug dropping events (F5 timeouts, retry_on_timeout workaround, upgrade ...
3	Aegis / Frostbyte AI - Identity Module Setup	Identity / SSO / MFA	Onboarding & deployment	The call is an onboarding kickoff for Aegis Identity with a setup timeline, Okta migration plan and next steps like API token setup and ...
4	Protect Team - Quarterly Planning	Backup / Protect	Internal planning	This is an internal Protect team Q2 quarterly planning call, covering roadmap scoping, capacity and tech debt, with backup and restore f...
5	Aegis / Harborview Banking - Threat Detection Tuning	Compliance & audit	Onboarding & deployment	Post-go-live Detect tuning session fixing configuration (alert mode on privileged access rules, alert routing, trusted location profiles...
6	Aegis / Meridian Capital - Service Reliability Discussion	Compliance & audit	Detect outage & fallout	The call centers on the six-hour Detect monitoring blackout, Ravi's admission of the single point of failure, remediation commitments, a...
7	ESCALATION: Northstar Pharma - Detect Outage Impact on Compliance	Compliance & audit	Detect outage & fallout	The call is an escalation about the ongoing Detect event pipeline outage, with lack of proactive notification, root cause, and update co...

What this shows. The LLM and the rules pick the same primary theme for about three calls in four, and the LLM's choice is almost always among the themes the rules also detected. So the LLM is rarely inventing anything, and is mostly **choosing which of several plausible themes matters most**.

The disagreements are where that judgement adds value. The largest group is calls about the **Comply v2 product launch** that the rules file under **Compliance & audit**, because the word "compliance" appears everywhere. The LLM separates the *product* from the *regulatory work*. Similarly, renewal calls dominated by the outage are labelled by what drove the conversation, not by the word "renewal".

8b. Certainty: where is the model unsure?

In [17]:
```
px.histogram(H, x="certainty", nbins=12, color="call_type", color_discrete_map=TYPE_COLORS,
             title="LLM certainty in the primary theme").update_layout(bargap=0.05)
```



In [18]:
```
H.sort_values("certainty")[["certainty", "primary_theme", "themes", "title", "rationale"]].head(6).reset_index(drop=True)
```

Out[18]:

	certainty	primary_theme	themes	title	rationale
0	0.40	Onboarding & deployment	[Compliance & audit, Onboarding & deployment]	Support Case #1750 - Vanta Health Systems SIEM Connector Issue	The call is a support case about a LogVault SIEM connector 3.4.1 bug dropping events (F5 timeouts, retry_on_timeout workaround, upgrade ...
1	0.40	Detect outage & fallout	[Compliance & audit, Detect outage & fallout, Comply v2 launch]	Aegis / Pinnacle Insurance - Detect Roadmap Review	The call is a Detect roadmap review, but the strongest driver is the March six-hour outage and the pipeline health visibility gap, with ...
2	0.50	Renewal & contracts	[Detect outage & fallout, Renewal & contracts, Internal planning, Competitive threat]	Customer Success - At-Risk Account Review	The internal at-risk account review centers on renewal outcomes (Brightpath two-year, Summit Trust one-year, Keystone add-on, Northstar ...
3	0.55	Onboarding & deployment	[Compliance & audit, Onboarding & deployment]	Aegis / Harborview Banking - Threat Detection Tuning	Post-go-live Detect tuning session fixing configuration (alert mode on privileged access rules, alert routing, trusted location profiles...
4	0.55	Renewal & contracts	[Compliance & audit, Detect outage & fallout, Comply v2 launch, Renewal & contracts, Competitive threat]	Aegis / Nova Retail Group - Renewal Discussion	The call is a pre-renewal account health discussion where the outage, SentinelShield evaluation, compliance reporting pain and Comply v2...
5	0.55	Comply v2 launch	[Compliance & audit, Comply v2 launch, Renewal & contracts, Competitive threat]	Support Case #4471 - Northstar Pharma Comply v2 HIPAA Report Export Failure	The call centers on a Comply v2 bug (error 5023, packages over 2 GB failing to split) breaking the HIPAA auditor export, though the audi...

What this shows. Certainty is high for most calls. The least certain calls are informative:

- **Genuinely mixed calls.** For example, a roadmap review dominated by the outage, or a renewal call where the outage, a competitor and pricing all matter. The multi-label `themes` field captures these properly even when the primary is a close call.
- **A gap in the taxonomy.** A LogVault SIEM-connector support case doesn't fit any theme well, and the model says so in its rationale. If integration issues grow, *Integrations / LogVault* should become an 11th theme. Low-certainty calls are a built-in signal for when to revise the taxonomy.

In production, calls below a certainty threshold (say 0.6) would go to a human review queue.

8c. Hand-labelled spot check

Agreement with rules isn't the same as being *right*. The final check is a human label on a sample. The cell below picks a stratified sample of 20 original calls (two per predicted theme). If no labels exist yet, it writes a template. Fill in `my_label` with one theme per call (without looking at the predictions), save it as `data/labels/topic_gold.csv`, and re-run the cell to get accuracy for both modes.

In [19]:
```
LABEL_DIR = ROOT / "data" / "labels"
GOLD = LABEL_DIR / "topic_gold.csv"

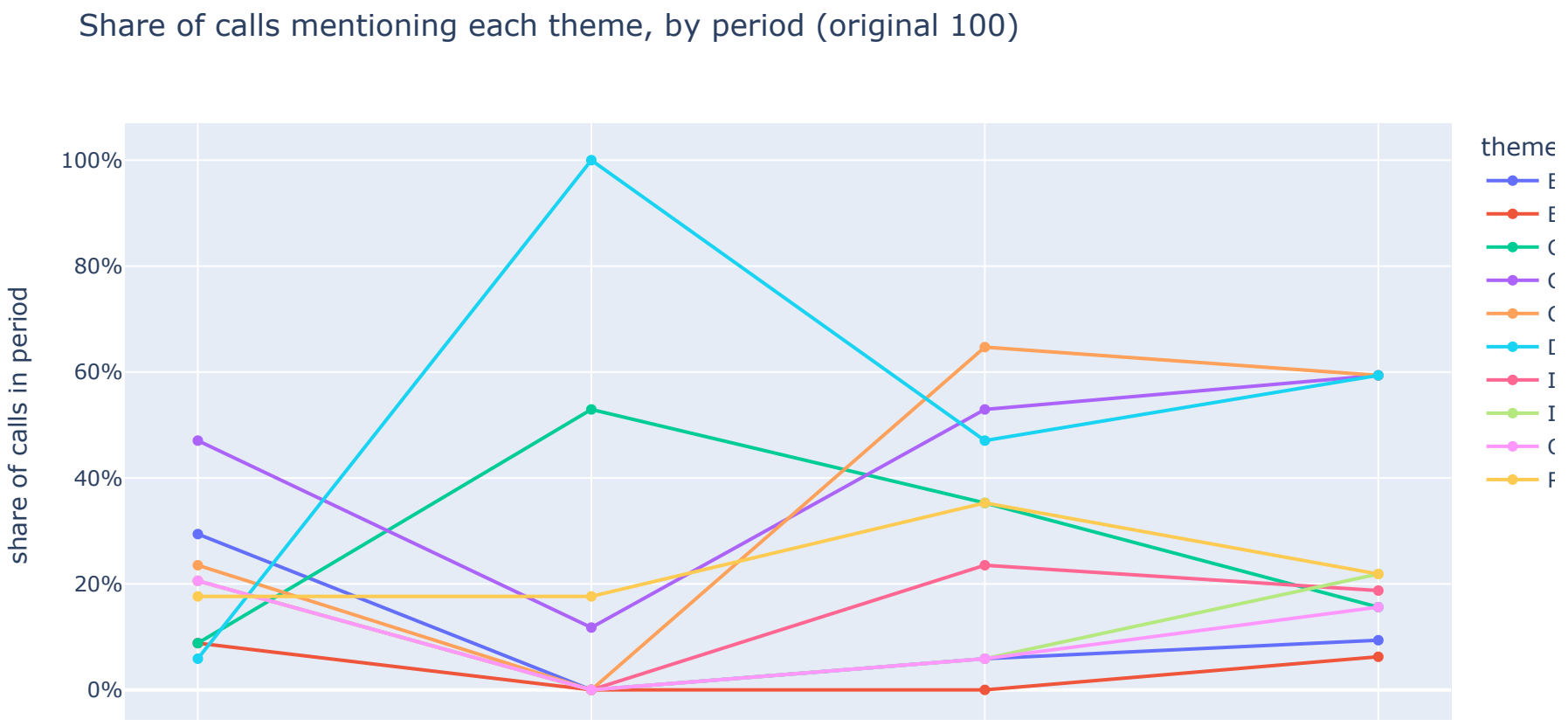
if GOLD.exists():
    gold = pd.read_csv(GOLD).dropna(subset=["my_label"])
    g = gold.merge(H[["call_id", "primary_theme", "themes"]], on="call_id") \
            .merge(R[["call_id", "primary_theme"]].rename(columns={"primary_theme": "rules_primary"}), on="call_id")
    print(f"{len(g)} hand-labelled calls")
    print(f"  Hybrid primary = my label: {(g['primary_theme'] == g['my_label']).mean():.0%}")
    print(f"  Hybrid themes include my label: {g.apply(lambda r: r['my_label'] in set(r['themes']), axis=1).mean():.0%}")
    print(f"  Rules primary = my label:  {(g['rules_primary'] == g['my_label']).mean():.0%}")
    display(g[g["primary_theme"] != g["my_label"]][["title", "my_label", "primary_theme", "rules_primary"]])
else:
    sample = (Ho.sample(frac=1, random_state=0)         # shuffle, then take two per predicted theme
                .groupby("primary_theme").head(2)
                .sample(frac=1, random_state=1))          # reshuffle so the order doesn't hint at the answer
    LABEL_DIR.mkdir(parents=True, exist_ok=True)
    tmpl = LABEL_DIR / "topic_gold_template.csv"
    sample[["call_id", "call_type", "title"]].assign(my_label="").to_csv(tmpl, index=False)
    print(f"No labels yet. Wrote a {len(sample)}-call template to {tmpl.relative_to(ROOT)}")
    print("Valid labels:", "; ".join(THEME_NAMES))
```

```
No labels yet. Wrote a 20-call template to data\labels\topic_gold_template.csv
Valid labels: Compliance & audit; Detect outage & fallout; Comply v2 launch; Renewal & contracts; Internal planning; Competitiv
e threat; Identity / SSO / MFA; Backup / Protect; Onboarding & deployment; Billing disputes
```

9. Themes over time

Because each call is dated, themes can be trended. That's the bridge from *what* calls are about to *how people feel* (`03_sentiment`).

In [20]:
```
tp = (Ho.assign(period=pd.Categorical(Ho["period"], PERIODS, ordered=True))
        .explode("themes").groupby(["period", "themes"], observed=False).size().rename("calls").reset_index())
tp["share"] = tp["calls"] / tp["period"].map(Ho["period"].value_counts())
px.line(tp, x="period", y="share", color="themes", markers=True,
        title="Share of calls mentioning each theme, by period (original 100)",
        labels={"share": "share of calls in period", "themes": "theme"}).update_yaxes(tickformat=".0%")
```



In [21]:
```
(Ho.groupby("primary_theme")
   .agg(calls=("call_id", "count"), avg_sentiment=("sentiment_score", "mean"))
   .sort_values("avg_sentiment").round(2))
```

Out[21]:

primary_theme	calls	avg_sentiment
Detect outage & fallout	23	2.32
Identity / SSO / MFA	9	2.74
Competitive threat	6	2.82
Backup / Protect	8	3.32
Billing disputes	4	3.45
Internal planning	11	3.88
Renewal & contracts	12	3.93
Compliance & audit	9	4.14
Onboarding & deployment	7	4.21
Comply v2 launch	11	4.50

What this shows. The themes follow the company's story, with two surprises.

- **The outage outlasts the incident.** Every call during the outage window touches it, as expected. But a month after full restoration it still comes up in more than half of all calls, *more* than during recovery, as renewals, audits and competitive evaluations bring it back up. For leadership, the cost of an incident is measured in months of conversations, not hours of downtime.
- **Competitive threat is outage-driven.** Competitor talk jumps from under 1 in 10 calls before the outage to about half of calls during it, then falls steadily after Comply v2 ships. That's consistent with the launch working as the "forward momentum" story the go-to-market team planned.
- **Comply v2** climbs during recovery (launch preparation) and stays high after GA.

Ranking themes by average sentiment shows the outage, Identity problems and competitive pressure as the most painful subjects. `03_sentiment` tests this with our own transcript-level scoring rather than relying on the vendor's score.

10. Does the pipeline handle new calls?

The synthetic transcripts were never seen when the taxonomy was designed. If they land in sensible themes, the pipeline generalises beyond the calls it was built on.

In [22]:
```
H[H["is_synthetic"]].sort_values("date")[["date", "call_type", "title", "primary_theme", "themes", "certainty"]] \
  .reset_index(drop=True)
```

Out[22]:

	date	call_type	title	primary_theme	themes	certainty
0	2026-05-06	external	Aegis / Brightpath Commerce - Detect Technical Session	Competitive threat	[Detect outage & fallout, Renewal & contracts, Competitive threat]	0.72
1	2026-05-08	support	Support Case #4471 - Northstar Pharma Comply v2 HIPAA Report Export Failure	Comply v2 launch	[Compliance & audit, Comply v2 launch, Renewal & contracts, Competitive threat]	0.55
2	2026-05-12	internal	Detect Chaos Engineering - Game Day Review	Detect outage & fallout	[Detect outage & fallout, Internal planning]	0.60
3	2026-05-14	external	Aegis / Summit Trust - Renewal Decision	Renewal & contracts	[Detect outage & fallout, Comply v2 launch, Renewal & contracts, Competitive threat, Identity / SSO / MFA]	0.62
4	2026-05-19	support	Support Case #8124 - Crestline Wealth Sync Fix Verification	Identity / SSO / MFA	[Identity / SSO / MFA]	0.95
5	2026-05-21	support	Support Case #5530 - Silverline Brands Protect Restore Test Failure	Backup / Protect	[Compliance & audit, Renewal & contracts, Competitive threat, Backup / Protect]	0.90
6	2026-05-27	external	Aegis / Keystone Health - Detect Add-on Decision	Renewal & contracts	[Compliance & audit, Detect outage & fallout, Comply v2 launch, Renewal & contracts, Onboarding & deployment]	0.62
7	2026-05-28	internal	Customer Success - At-Risk Account Review	Renewal & contracts	[Detect outage & fallout, Renewal & contracts, Internal planning, Competitive threat]	0.50
8	2026-06-02	support	Support Case #6018 - Frostbyte AI Contractor Attribute Mapping Rejected	Identity / SSO / MFA	[Compliance & audit, Identity / SSO / MFA]	0.90
9	2026-06-04	external	Aegis / Vanta Health Systems - Q2 Business Review	Backup / Protect	[Detect outage & fallout, Backup / Protect]	0.85
10	2026-06-09	internal	Pricing & Packaging Review - Comply Usage Tiers	Internal planning	[Comply v2 launch, Renewal & contracts, Internal planning, Competitive threat, Billing disputes]	0.55
11	2026-06-11	support	Support Case #7311 - Axiom Labs Elastic Seat Invoice Error	Billing disputes	[Renewal & contracts, Billing disputes]	0.96

	date	call_type	title	primary_theme	themes	certainty
12	2026-06-16	internal	Protect Scheduler Rewrite - Design Review	Backup / Protect	[Internal planning, Backup / Protect]	0.80
13	2026-06-24	internal	Q2 Executive Business Review	Internal planning	[Detect outage & fallout, Renewal & contracts, Internal planning]	0.85
14	2026-07-01	external	Aegis / Larkspur Financial - Onboarding Kickoff	Onboarding & deployment	[Compliance & audit, Comply v2 launch, Onboarding & deployment]	0.88

11. Summary

Question	Answer
Categories	10 themes, from Detect outage & fallout through Billing disputes, each with a definition and a keyword rule
Approach	Hybrid. Clustering discovers, the taxonomy defines, Claude classifies (multi-label + certainty + rationale), rules provide a fallback and a check
Why	Provided topics are too fragmented; clustering alone forms groups around customer names and can't multi-label; rules alone match words, not meaning
How much to trust it	High agreement with independent rules; disagreements are explainable judgement calls; low-certainty calls flag taxonomy gaps; hand-label check in 8c
Cost	Fractions of a cent per call, cached so re-runs are free
Generalises	New (synthetic) calls are classified sensibly without changes

Limitations. The taxonomy was designed on this dataset and would need periodic review (8b shows how to spot when). Certainty is the model's own estimate, not a calibrated probability. The hand-labelled sample is small.

Next: `03_sentiment.ipynb`